DEPARTMENT OF ORAL BIOLOGY
DR. ISHRAT UL EBAD KHAN INSTITUTE OF ORAL HEALTH SCIENCES
PROBLEM BASED LEARNING SESSION KEY DATED 23-10-2026

Scenario 1: Oral Mucosa, Tooth Development and Wear

1. Buccal mucosa versus hard palate

- **Buccal mucosa:** Non-keratinized lining mucosa; flexible and mobile.
- **Hard palate:** Keratinized/parakeratinized masticatory mucosa; firmly attached and friction-resistant.

2. Response to stimuli

- Buccal mucosa is less resistant to friction.
- Hard palate withstands chewing forces.
- Both detect thermal stimuli; burning requires further assessment.

3. Tissue exposed by recession

- Cementum initially; dentin if cementum is lost.

4. CEJ and its relationships

- Junction of crown enamel and root cementum.
- Cementum overlaps enamel, meets it edge-to-edge, or leaves a gap exposing dentin.

5. Recession and sulcus depth

- Recession: Apical displacement of the gingival margin.
- Sulcus depth does not necessarily increase.

6. Tissue-forming cells

- Dentin: **Odontoblasts.**
- Enamel: **Ameloblasts.**
- Cementum: **Cementoblasts.**

7. Types of tooth wear

- **Attrition:** Tooth-to-tooth contact; matching wear facets.
- **Abrasion:** External mechanical action, such as brushing.
- **Erosion:** Non-bacterial acids dissolve tooth tissue.

8. Dentin hypersensitivity

- Brief, sharp pain from exposed dentin after excluding other causes.
- Stimulus → dentinal fluid movement → activation of pulpal nerves, mainly **A-delta fibres.**

PREPARED BY: DR. MISBAH ASHRAF MOTEN

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Scenario 2: Premolar Identification and Development

1. Identification

- **Three-cusp mandibular second premolar**; a normal anatomical variant.

2. Identifying features

- One buccal and two lingual cusps.
- Y-shaped groove pattern and broad occlusal table.

3. Comparison with a canine

- **Premolar:** Multiple cusps; crushing and grinding.
- **Canine:** Single cusp; tearing and guiding occlusion.

4. Crown calcification

- Begins: **2–2½ years.**
- Completed: **6–7 years.**

5. Root completion

- Approximately **13–14 years.**

6. Root development

- Begins after crown formation is substantially complete.
- **HERS** guides root development; its epithelial diaphragm helps determine root number.

7. Roots and canals

- Usually **one root and one canal**; variations occur.

8. Agenesis

- Among the commonly missing teeth, especially when third molars are excluded.

Frequency and ranking vary by population.

PREPARED BY: DR. SABA ARSHAD

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PBL Scenario 1 Rubric — Total 10 Marks

Assessment criterion	Expected performance	Marks
Oral mucosa	Compares buccal mucosa and hard palate (1); Explains their responses to stimuli (1).	2
Recession and CEJ	Identifies exposed cementum/dentin (0.5); Defines CEJ and its three relationships (1); Distinguishes recession from sulcus depth (0.5).	2
Tissue-forming cells	Correctly identify odontoblasts, ameloblasts and cementoblast.	1.5
Tooth wear	Differentiates attrition, abrasion and erosion (1.5); Identifies flat incisal wear as consistent with attrition (0.5).	2
Dentin hypersensitivity	Defines hypersensitivity (1); Explains dentinal fluid movement and nerve activation (1).	2
Participation	Contribute actively and participation in teamwork.	0.5
Total		**10**

Prepared by Dr. Misbah Ashraf Moten

Scoring: Award full marks for accurate, complete responses; partial marks for partly correct responses; zero for incorrect or absent responses. Accept equivalent terminology and reasonable textbook variations in developmental ages.

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PBL Scenario 2 Rubric — Total 10 Marks

Assessment criterion	Expected performance	Marks
Tooth identification	Identifies the mandibular second premolar (1); Recognize the three-cusp form as a normal variant (1).	2
Identifying features	Describes one buccal and two lingual cusps (1); Identifies the Y-shaped groove pattern (0.5).	1.5
Comparison with canine	Compares cusp number, crown form and function.	1
Developmental chronology	States crown calcification begins at 2–2½ years (0.5), Crown completion at 6–7 years (0.5), Root completion at 13–14 years (0.5).	1.5
Root development	Explains the role of HERS and the epithelial diaphragm.	1
Roots, pulp canals and anatomical variations	States the usual single root and canal, acknowledging variation (1); Recognize frequent agenesis (0.5).	1.5
Occlusal anatomy	Explain the occlusal anatomy.	1
Participation	Contribute actively and participation in teamwork.	0.5
Total		**10**

Prepared by Dr. Saba Arshad

Scoring: Award full marks for accurate, complete responses; partial marks for partly correct responses; zero for incorrect or absent responses. Accept equivalent terminology and reasonable textbook variations in developmental ages.

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"A 58-year-old man and long-term smoker wears a partial upper denture. He complains of a burning sensation on the palate and irritation along the inner lining of his cheek. On examination, the buccal mucosa irritated in some areas. The patient reports discomfort when consuming hot or spicy foods. He also presents with gum recession on the lower premolars, exposing yellow root surfaces. The biting edges of his lower incisors are worn flat, with small, shiny yellow areas, and two lower incisors show slight mobility."

Section A: Oral Histology and Mucosa

1. Compare the histological features of buccal mucosa and hard palate.
2. Explain why the two regions respond differently?
3. Due to gingival recession, which structural loss would be observed at the cervix of the tooth.

Section B: Tooth Development and Anatomy

3. What is the cemento-enamel junction, and what are the three arrangments of Cemento-enamel junction?
4. How much loss of gingival crevicular depth can be associated with gingival recession
5. Identify the cells responsible for dentin, enamel and cementum formation.

Section C: Tooth Wear and Dentin Hypersensitivity

5. Differentiate attrition, abrasion, and erosion using their causes and clinical features.
6. Define dentin hypersensitivity and explain its mechanism using the hydrodynamic theory.

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"During a supervised clinical examination, a 13-year-old patient presents with a lower premolar looking abnormal because it has more than two raised areas, making Y pattern on the occlusal surface. There is no reported pain. A student says, 'All premolars are bicuspids, so the extra cusp must be abnormal.' Your group must decide whether that statement is justified, what to examine next, and how to explain the anatomy."

Section A: Tooth Identification

1. Does this establish an anomaly? Identify the premolar?
2. How did you identify the tooth based on classic features.
3. Compare the tooth with canines.

Section B: Development and Chronology

4. At what time point, crown formation is initiated and completed during development.
5. At what time point, root formation is completed during development.
6. Describe the developmental stage and embryological structure that defines the number of roots in this tooth during formation.

Section C: Anatomical variations and root canal morphology

6. How many roots and canals are commonly present in this tooth?
7. Order the tooth among the highly prevalent anatomical variations related to missing teeth in the oral cavity

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